



Islamic University of Technology

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Course: PHY - 4241

Assignment No: 03

Assignment Title: CHAPTER - 09  
CHAPTER - 10

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Section:01

Dept. of Computer Science and Engineering (CSE)

## Chapter-9:

25. Find  $i(t)$  in the following equations using phasors:

$$(a) \ 2 \frac{di}{dt} + 3i(t) = 4 \cos(2t - 45^\circ)$$

$$(b) \ 10 \int i dt + \frac{di}{dt} + 6i(t) = 5 \cos(5t + 22^\circ) \text{ A.}$$

Solution:

$$(a) \ 2j\omega I + 3I = 4 \angle -45^\circ$$

Comparing with standard wave equation:  
 $\omega = 2$

$$\Rightarrow 2j \times 2 \times I + 3I = 4 \angle -45^\circ$$

$$\Rightarrow I(j4 + 3) = 4 \angle -45^\circ$$

$$\Rightarrow I = \frac{4 \angle -45^\circ}{4j + 3}$$

$$= \frac{4 \angle -45^\circ}{5 \angle 53.13^\circ}$$

$$\therefore I = 0.8 \angle -98.13^\circ$$

$$\therefore i(t) = 0.8 \cos(2t - 98.13^\circ)$$

$$(b) 10 \frac{I}{j\omega} + j\omega I + 6I = 5 \angle 22^\circ$$

Comparing with standard wave equation:  
 $\omega = 5$ .

$$\Rightarrow 10 \frac{I}{j \times 5} + j \times 5 I + 6I = 5 \angle 22^\circ$$

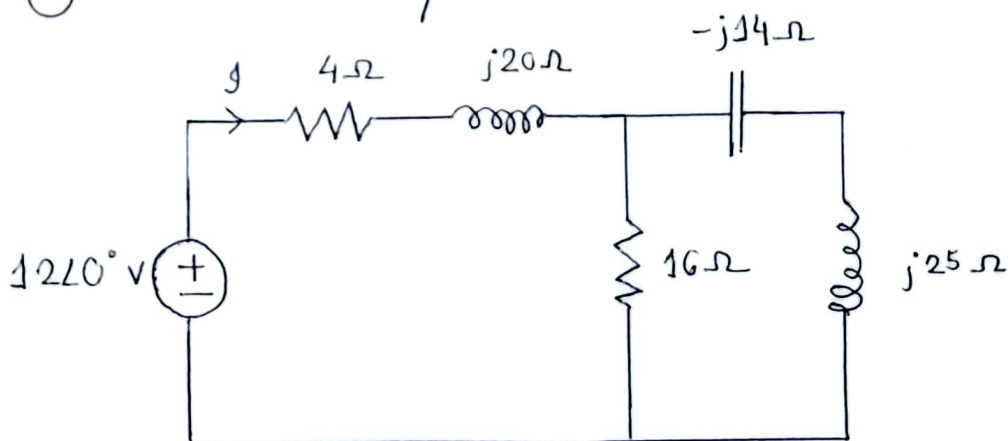
$$\Rightarrow I \left( \frac{2}{j} + j5 + 6 \right) = 5 \angle 22^\circ$$

$$\Rightarrow I = \frac{5 \angle 22^\circ}{6.708 \angle 26.56^\circ}$$

$$\therefore I = 0.745 \angle -4.56^\circ$$

$$\therefore i(t) = 0.745 \cos(5t - 4.56^\circ)$$

39. For the circuit shown in the figure, find  $Z_{eq}$  and use that to find current  $I$ .  
 Given,  $\omega = 10 \text{ rad/s}$



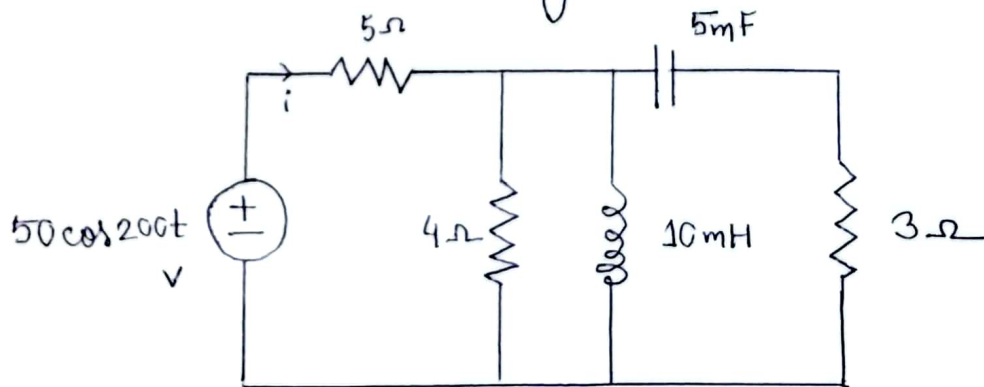
Solution:

$$\begin{aligned} Z_{eq} &= 4 + 20j + 10 \parallel (-14j + 25j) \\ &= 4 + 20j + 10 \parallel 11j \\ &= 4 + 20j + 5.47 + 4.9j \\ &= 9.475 + j24.977 \Omega \end{aligned}$$

$$\therefore I = \frac{V}{Z_{eq}} = \frac{12}{9.475 + j24.977} = 0.449 \angle -69.23^\circ$$

$$i(t) = 0.449 \cos(10t - 69.23^\circ) \text{ A}$$

44 Calculate  $i(t)$  in the given circuit.



Solution:

Comparing  $50 \cos 200t$  v with the standard equation, we get:

$$\omega = 200$$

10 mH can be written as:

$$j\omega L = j \times 200 \times 10 \times 10^{-3} = j2$$

5 mF can be written as:

$$\frac{1}{j\omega C} = \frac{1}{j \times 200 \times 5 \times 10^{-3}} = -j$$

We have  $4\Omega$ ,  $5\text{mF}$ ,  $10\text{mH}$  and  $3\Omega$  in parallel connection.

$\therefore$  We can write:

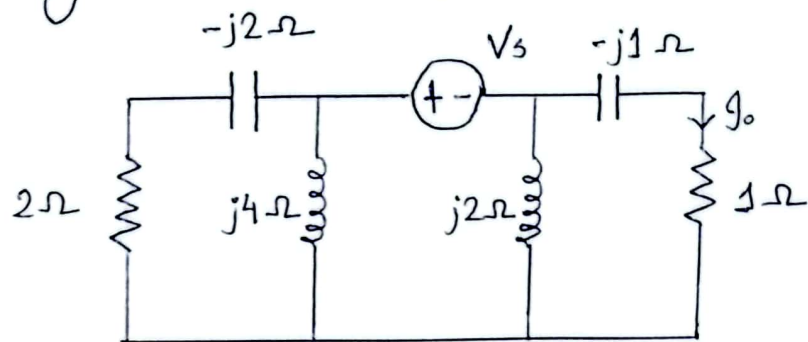
$$\begin{aligned} Z &= \left( \frac{1}{4} + \frac{1}{j2} + \frac{1}{3-j} \right)^{-1} \\ &= 1.1892 + 0.865j \end{aligned}$$

$$\therefore I(t) = \frac{50 \angle 0^\circ}{5 + 1.1892 + 0.865j}$$

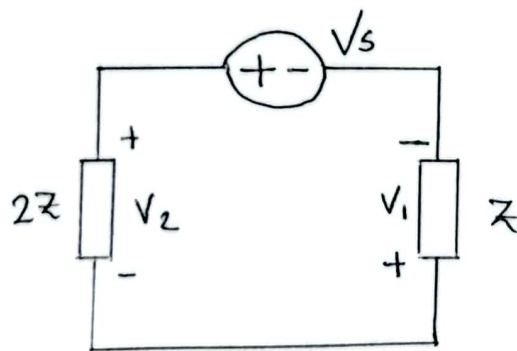
$$= 8.001 \angle -7.956^\circ$$

$$\therefore i(t) = 8.001 \cos(200t - 7.956^\circ) \text{ A.}$$

54. In the given circuit, find  $V_s$  if  $I_o = 2 \angle 0^\circ \text{ A}$ .



Solution: Since the left portion of the circuit is twice of that of the right portion, the equivalent circuit results in:



$$V_1 = I_o (1-j) = 2(1-j)$$

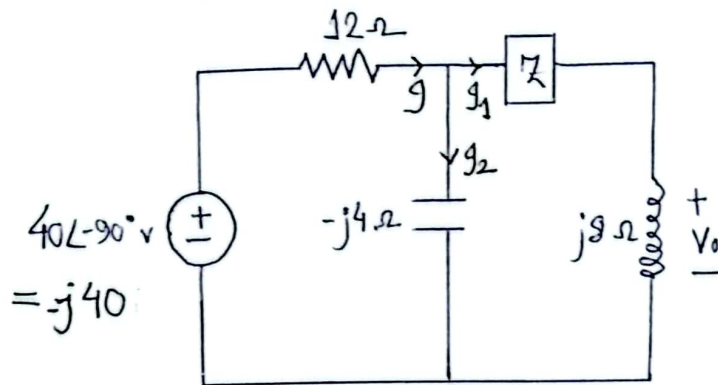
$$V_2 = 2V_1 = 4(1-j)$$

$$\therefore V_s = -V_1 - V_2 = -2 + j2 - 4 + j4 = -6(1-j)$$

$$\Rightarrow V_s = 8.485 \angle 135^\circ \text{ volts}$$



55. Find  $Z$  in the network of the given figure where,  $V_o = 8 \angle 0^\circ \text{ V}$



Solution:

$$g_1 = \frac{V_o}{j8} = \frac{8}{j8} = -j$$

$$g_2 = \frac{g_1(Z + j8)}{-j4} = \frac{(-j)(Z + j8)}{-j4} = \frac{Z}{8} + j2$$

$$\begin{aligned} \therefore g &= g_1 + g_2 \\ &= -j + \frac{Z}{8} + j2 \\ &= \frac{Z}{8} + j \end{aligned}$$

We can write:

$$-j40 = 12g + g_1(Z + j8)$$

$$\Rightarrow -j40 = 12\left(\frac{Z}{8} + j\right) + (-j)(Z + j8)$$

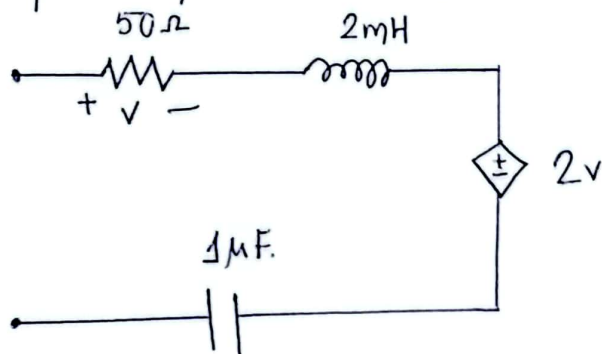
$$\Rightarrow -j40 - 1.5Z - j12 + jZ - 8 = 0$$

$$\Rightarrow Z(j - 1.5) - j52 - 8 = 0$$

$$\Rightarrow Z = \frac{8 + j52}{j - 1.5} = \frac{52.62 \angle 81.25^\circ}{1.803 \angle 146.31^\circ} = 29.18 \angle -65.06^\circ$$

$$\therefore Z = 12.304 - j26.46 \, \Omega$$

62. Find the input impedance  $Z_{in}$  at  $10 \text{ krad/s}$ .

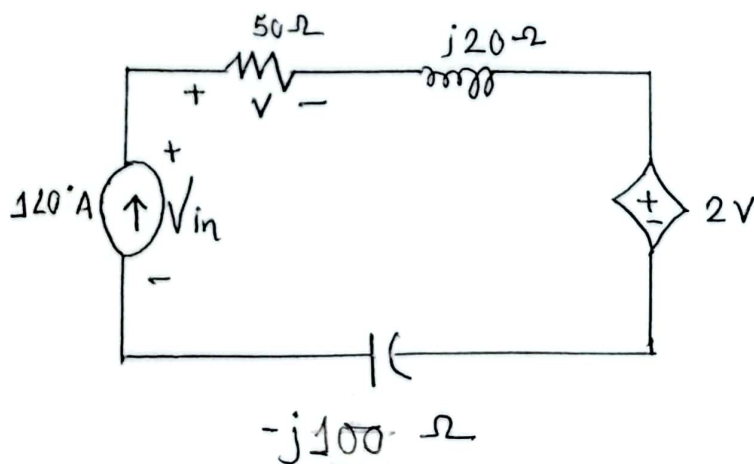


Solution:

$2 \text{ mH}$  can be written as:

$$j\omega L = j \times (10 \times 10^3) (2 \times 10^{-3}) = j20$$

$1 \mu\text{F}$  can be written as:  $\frac{1}{j\omega C} = \frac{1}{j \times (10 \times 10^3) (10^{-6})} = -j100$



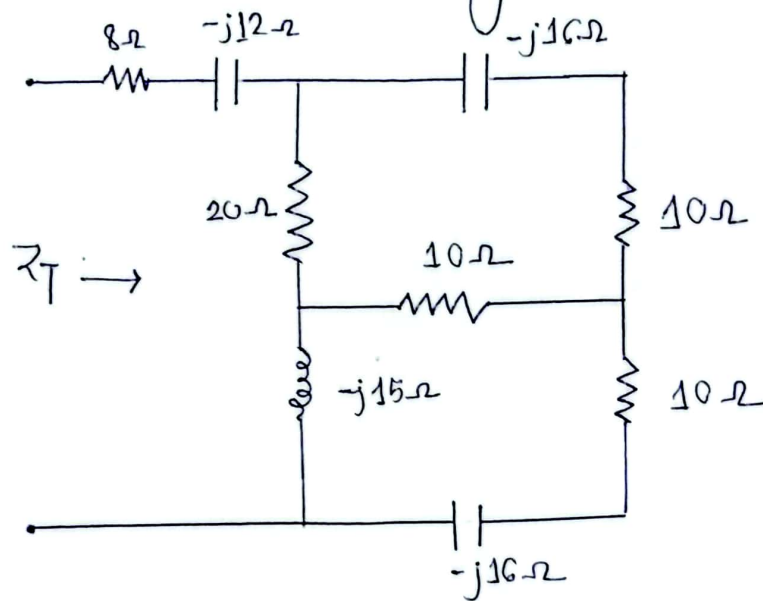
$$V = (120^\circ)(50) = 50$$

$$V_{in} = (120^\circ)(50 + j20 - j100) + (2 \times 50) = 150 - 80j$$

$$\therefore Z_{in} = \frac{V_{in}}{120^\circ} = \frac{150 - j80}{120^\circ} = 150 - j80 \Omega$$



63. Find the value of  $Z_T$  in the given circuit.



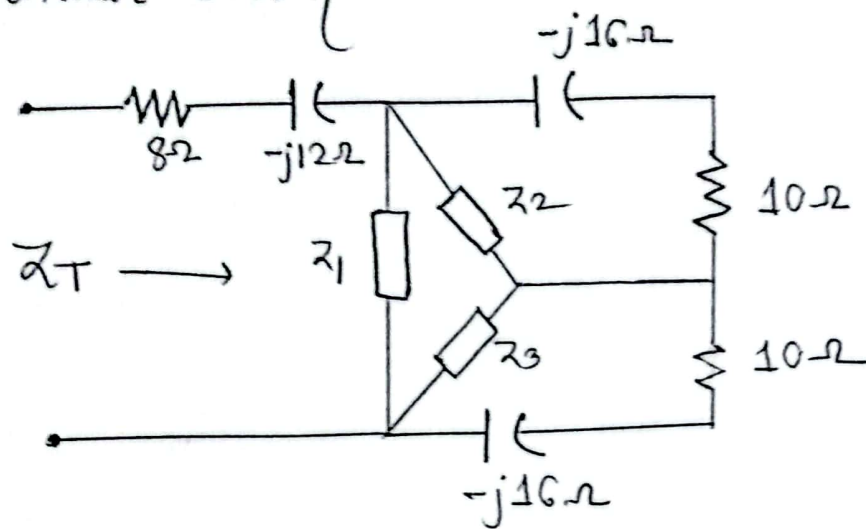
Solution: Initially, we replace the  $\pi$  circuit configuration composed of  $20\Omega$ ,  $10\Omega$  and  $j15\Omega$  impedances with corresponding delta

$$Z_1 = \frac{200 + j150 + j300}{40} = 20 + j15$$

$$Z_2 = \frac{200 + j150 + j300}{j15} = 30 - j13.33$$

$$Z_3 = \frac{200 + j150 + j300}{20} = 10 + j22.5$$

The new circuit becomes:



Then,

$$Z_2 \parallel (10 - j16) = 8.72 - j8.93$$

$$Z_3 \parallel (10 - j16) = 21.76 - j3.821$$

$$\therefore Z_T = 8 - j12 + Z_1 \parallel (8.72 - j8.93 + 21.76 - j3.821)$$

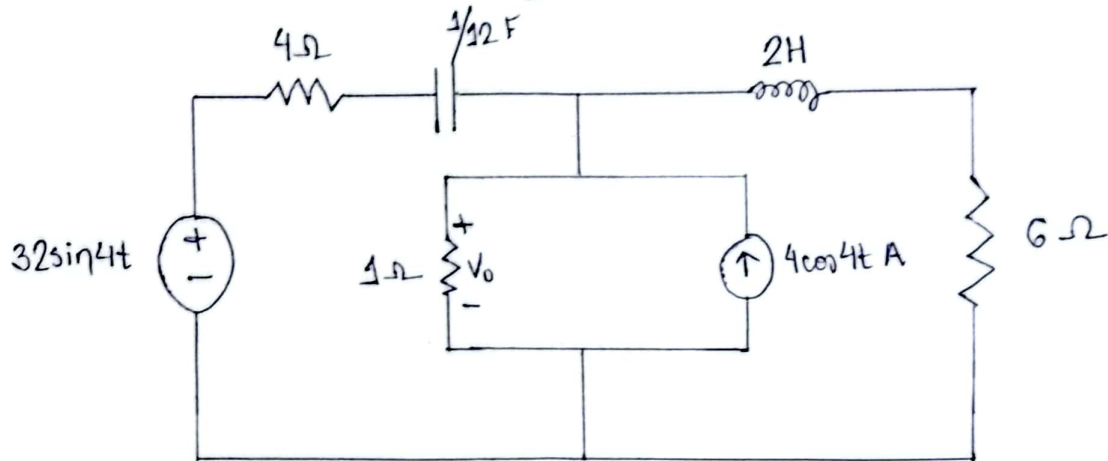
$$= 8 - j12 + Z_1 \parallel (30.48 - j12.75)$$

$$= 34.68 - j6.93 \Omega$$

Ans.

### Chapter-10:

3. Determine  $v_o$  in the given circuit.



Solution: From the figure, we can write:  
 $\omega = 4 \text{ rad/s.}$

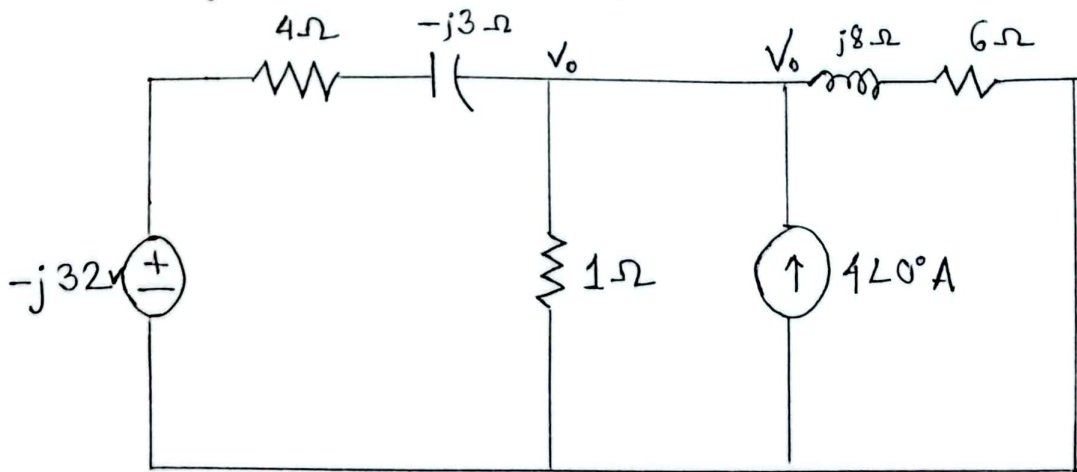
$4\cos 4t$  can be written as  $4\angle 0^\circ$ .

$32\sin 4t$  can be written as  $32\angle -90^\circ = -j32$ .

$2H$  can be written as  $j\omega L = j \times 4 \times 2 = j8$ .

$1/12 F$  can be written as  $\frac{1}{j\omega C} = \frac{1}{j \times 4 \times 1/12} = -j3$ .

The circuit becomes:



Applying nodal analysis, we get:

$$\frac{-j32 - V_0}{4 - j3} + 4 = \frac{V_0}{1} + \frac{V_0}{6 + j8}$$

$$\Rightarrow \frac{-j32}{4 - j3} + 4 = \left(1 + \frac{1}{4 - j3} + \frac{1}{6 + j8}\right)V_0$$

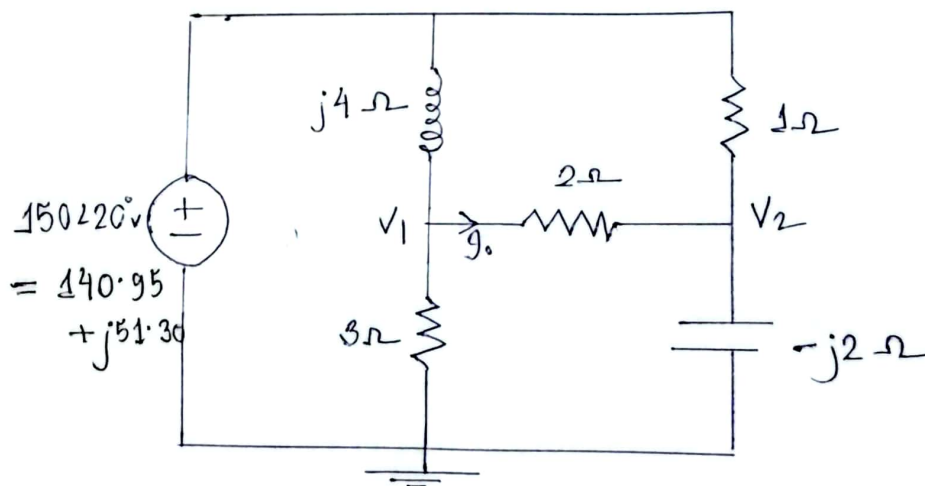
$$\Rightarrow V_0 = \frac{7.84 - j5.12}{1.22 + j0.04}$$

$$= \frac{9.364 \angle -33.15^\circ}{1.2207 \angle 1.88^\circ}$$

$$\therefore V_0 = 7.67 \angle -35.03^\circ$$

$\therefore$  the value of  $v_0$  is,  $v_0(t) = 7.67 \cos(4t - 35.03^\circ)$

17. By nodal analysis, obtain current  $i_0$  in the circuit



Solution: At node -01, we find:

$$\frac{140.95 + j51.3 - V_1}{j4} = \frac{V_1}{3} + \frac{V_1 - V_2}{2}$$

$$\Rightarrow (6 - j20)V_1 + j12V_2 = 845.7 + j307.8 \quad \text{--- (i)}$$

At node -02, we find:

$$\frac{140.95 + j51.3 - V_2}{1} + \frac{V_1 - V_2}{2} = \frac{V_2}{-j2}$$

$$\Rightarrow 2V_1 - (6 + j)V_2 = 563.8 + j205.2 \quad \text{--- (ii)}$$

$$\Rightarrow 3V_1 - (9 + j1.5)V_2 = 845.7 + j307.8$$

Representing (i) and (ii) as matrices, we get:

$$\begin{bmatrix} 845.7 + j307.8 \\ 845.7 + j307.8 \end{bmatrix} = \begin{bmatrix} 6 - j20 & j12 \\ 3 & -(9 + j1.5) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$\therefore \Delta = \begin{vmatrix} 6 - j20 & j12 \\ 3 & -(9 + j1.5) \end{vmatrix} = -84 + j135.$$

$$\Delta_1 = \begin{vmatrix} 845.7 + j307.8 & j12 \\ 845.7 + j307.8 & -(9 + j1.5) \end{vmatrix} = -3456 - j14187.15$$

$$\Delta_2 = \begin{vmatrix} 6 - j20 & 845.7 + j307.8 \\ 3 & 845.7 + j307.8 \end{vmatrix} = 8693.1 - j15990.6$$



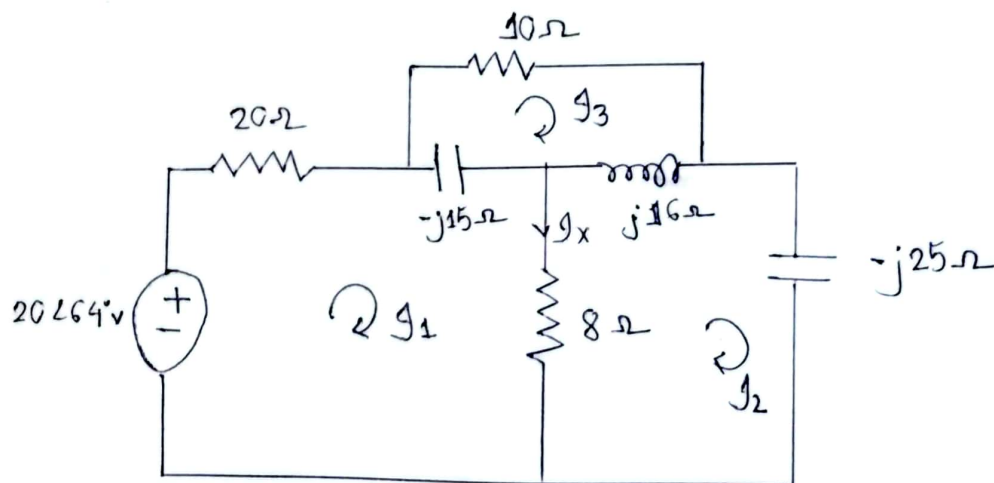
$$\therefore V_1 = \frac{\Delta_1}{\Delta} = 91.84 \angle 134.42^\circ$$

$$V_2 = \frac{\Delta_2}{\Delta} = 114.47 \angle 176.64^\circ$$

$$\therefore I_0 = \frac{V_1 - V_2}{2} = \frac{\Delta_1 - \Delta_2}{2\Delta} = \frac{-12149.1 + j1803.45}{-168 + j270}$$

$$\Rightarrow I_0 = 38.62 \angle 49.67^\circ \text{ A.}$$

39. Find the values of  $I_1, I_2, I_3$  and  $I_x$ . Ans.



Solution: For mesh-01:

$$(28 - j15)I_1 - 8I_2 + j15I_3 = 20\angle 64^\circ \quad \text{--- (i)}$$

$$\text{For mesh-02: } -8I_1 + (8 - j9)I_2 - j16I_3 = 0 \quad \text{--- (ii)}$$

$$\text{For mesh-03: } j15I_1 - j16I_2 + (10 + j)I_3 = 0 \quad \text{--- (iii)}$$



Representing eq<sup>n</sup> (i) (ii) and (iii) in matrix form:

$$\begin{bmatrix} 28-j15 & -8 & j15 \\ -8 & 8-j9 & -j16 \\ j15 & -j16 & 10+j \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 20 \angle 64^\circ \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore \Delta = \begin{vmatrix} 28-j15 & -8 & j15 \\ -8 & 8-j9 & -j16 \\ j15 & -j16 & 10+j \end{vmatrix} = 5750 - j9560.$$

$$\Delta_1 = \begin{vmatrix} 20 \angle 64^\circ & -8 & j15 \\ 0 & 8-j9 & -j16 \\ 0 & -j16 & 10+j \end{vmatrix} = 4498.78 + j5482.75$$

$$\Delta_2 = \begin{vmatrix} 28-j15 & 20 \angle 64^\circ & j15 \\ -8 & 0 & -j16 \\ j15 & 0 & 10+j \end{vmatrix} = -2661.77 - j5822.42$$

$$\Delta_3 = \begin{vmatrix} 28-j15 & -8 & 20 \angle 64^\circ \\ -8 & 8-j9 & 0 \\ j15 & -j16 & 0 \end{vmatrix} = -1327.41 - j2356.6.$$

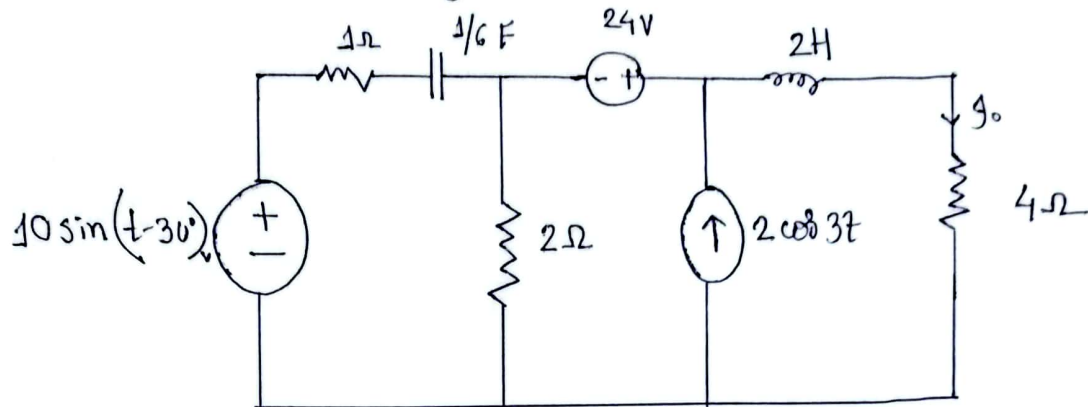
$$\therefore I_1 = \frac{\Delta_1}{\Delta} = 0.636 \angle 109.60^\circ \text{ A.}$$

$$I_2 = \frac{\Delta_2}{\Delta} = 0.574 \angle -55.59^\circ \text{ A.}$$

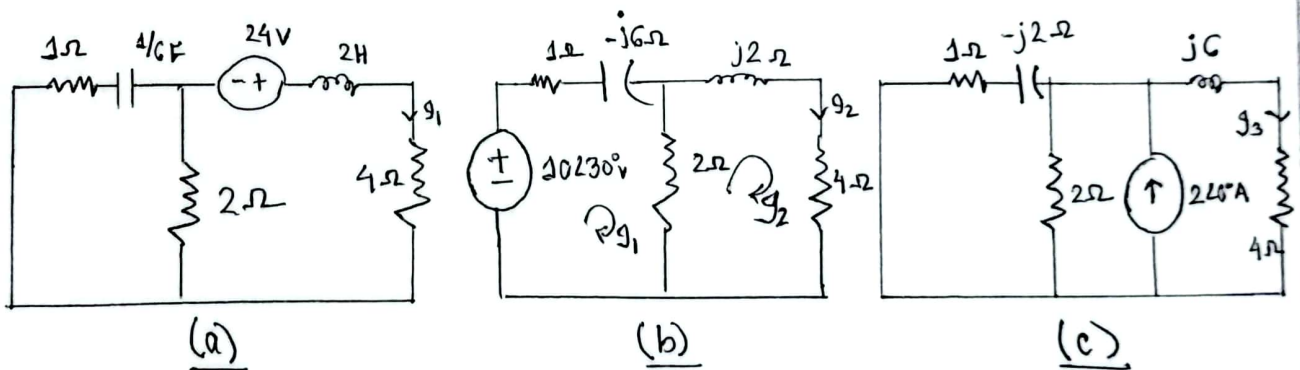
$$I_3 = \frac{\Delta_3}{\Delta} = 0.242 \angle -60.42^\circ \text{ A}$$

$$\text{and } I_x = I_1 - I_2 = 1.19 \angle 116.62^\circ \text{ A} \quad \text{Ans.}$$

47. Determine  $I_0$  using the superposition principle.



Solution: Let,  $I_0 = I_1 + I_2 + I_3$  where each of the currents are respectively due to 24V dc source, ac voltage source and ac current source.



For  $I_1$ , figure (a) is considered:

$$I_1 = \frac{24}{2+4} = 4 \text{ A.}$$

For  $I_2$ , figure (b) is considered:

$$W = 1.$$

$$2\text{H} \rightarrow j\omega L = j \times 1 \times 2 = j2$$

$$\frac{1}{6}\text{F} \rightarrow \frac{1}{j\omega C} = \frac{1}{j \times 1 \times \frac{1}{6}} = -j6$$

For mesh-01:

$$-10\angle 30^\circ + (3-j6)I_1 - 2I_2 = 0$$

$$\Rightarrow 10\angle 30^\circ = 3(1-j2)I_1 - 2I_2 \text{ ————— (i)}$$

For mesh-02:

$$0 = -2I_1 + (6+j2)I_2$$

$$\Rightarrow I_1 = (3+j)I_2 \text{ ————— (ii)}$$

From (i) and (ii)  $\Rightarrow$

$$10\angle 30^\circ = 15 - j15 I_2$$

$$\Rightarrow I_2 = 0.538\angle -51.74^\circ$$

Hence,  $I_2 = 0.538 \cos(1 - 51.74^\circ) \text{ A.}$

For  $I_3$ , figure (c) is considered.

$$n = 3.$$

$$2H \rightarrow j\omega L = j \times 3 \times 2 = j6$$

$$\frac{1}{6}F \rightarrow \frac{1}{j\omega C} = \frac{1}{j \times 3 \times \frac{1}{6}} = -j2$$

$$\therefore 2 \parallel (1-j2) = \frac{14}{13} - j\frac{8}{13}$$

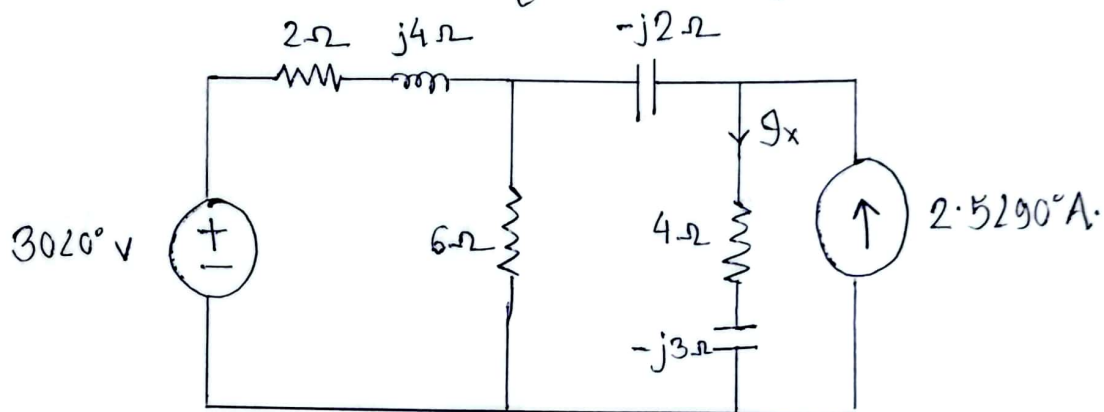
Using current division, we can write:

$$I_3 = \frac{\left(\frac{14}{13} - j\frac{8}{13}\right)(2\angle 0^\circ)}{4 + j6 + \frac{14}{13} - j\frac{8}{13}} = 0.3352 \angle -76.43^\circ$$

Hence,  $I_3 = 0.3352 \cos(3t - 76.43^\circ) \text{ A}$ .

$$\therefore I_0 = 4 + 0.538 \cos(1 - 51.74^\circ) + 0.3352 \cos(3t - 76.43^\circ)$$

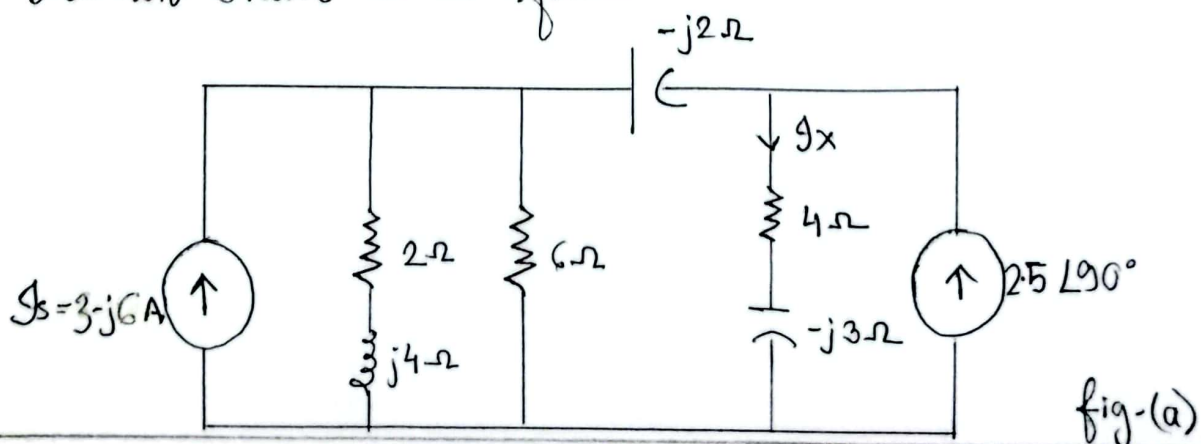
52. Use source transformation to find  $I_x$  Ans.



Solution: We transform the voltage source to a current source:

$$I_s = \frac{30\angle 0^\circ}{2 + j4} = 3 - j6$$

The new circuit is as follows:

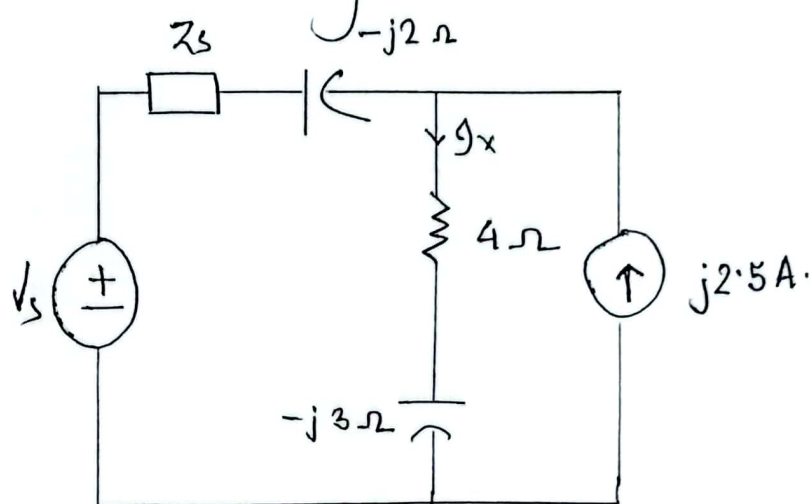




Let,  $Z_s = 6 \parallel (2 + j4) = 2.4 + j1.8$

$V_s = I_s Z_s = (3 - j6)(2.4 + j1.8) = 9(2 - j)$

With these, we transform the current source on left side to a voltage source and obtain fig (b)

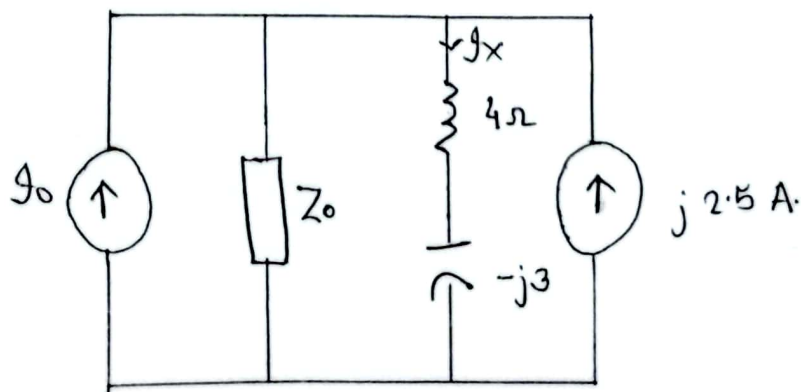


(b).

Let,  $Z_0 = Z_s - j2 = 2.4 - j0.2$

$I_0 = \frac{V_s}{Z_0} = \frac{9(2 - j)}{2.4 - j0.2} = 7.76 - j3.10$

With these, we transform the voltage source to a current source and obtain fig (c).





Using current division,

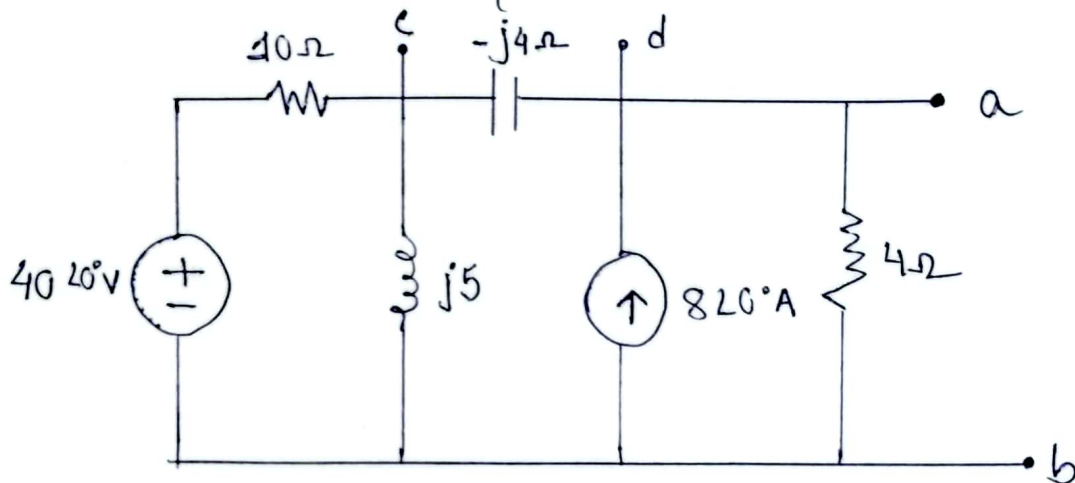
$$I_x = \frac{Z_0}{Z_0 + 4 - j3} (I_0 + 2.5j) = 2.62 \angle 17.38^\circ \text{ A}$$

60. Find Thevenine equivalent in

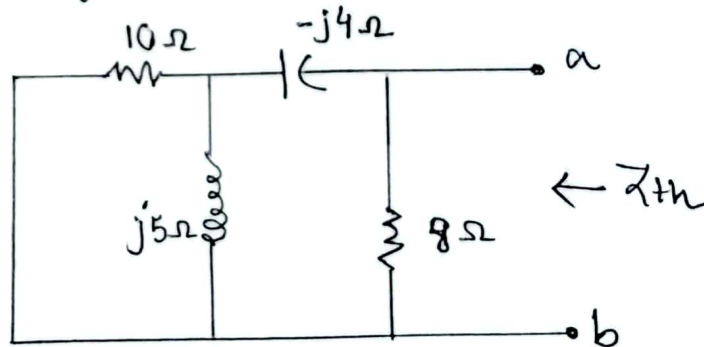
Ans.

(a) terminals a-b

(b) terminals c-d.

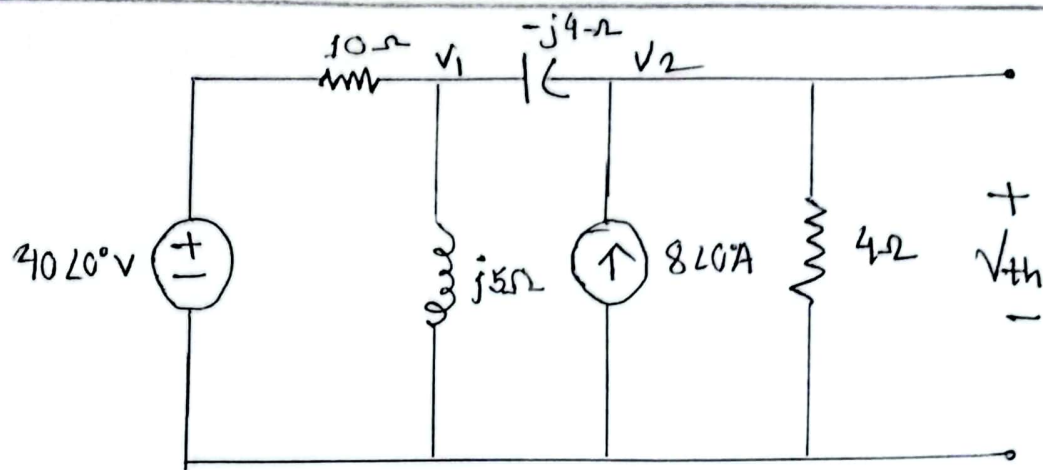


Solution: (a) To find  $Z_{th}$ , the circuit is as follows:



$$Z_{th} = 8 \parallel (-j4 + 10 \parallel j5) = 1.6 \Omega$$

To find  $V_{th}$ , we construct the circuit as follows:



At node - 01:

$$\frac{40 - V_1}{10} = \frac{V_1}{j5} + \frac{V_1 - V_2}{-j4}$$

$$\Rightarrow (j+2)V_1 - j5V_2 = 40 \quad \text{--- (i)}$$

At node - 02:

$$8 + \frac{V_1 - V_2}{-j4} = \frac{V_2}{4}$$

$$\Rightarrow V_1 = (1-j)V_2 + j32 \quad \text{--- (ii)}$$

Substituting (ii) in to (i):

$$(j+2)\{(1-j)V_2 + j32\} - j5V_2 = 40$$

$$\Rightarrow 3V_2 - jV_2 - 32 - j5V_2 = 40 - j64$$

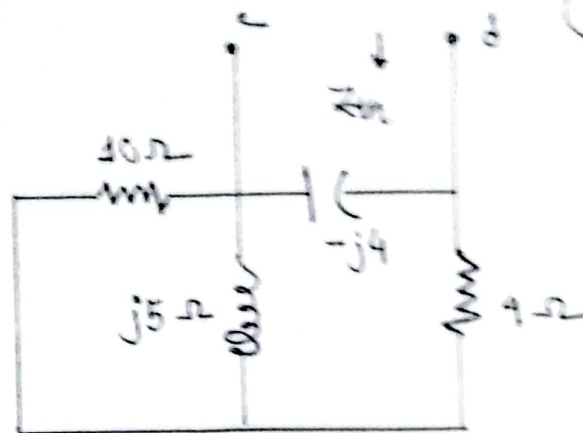
$$\Rightarrow V_2 = \frac{40 + 32 - j64}{3 + (-j) - j5}$$

$$\therefore V_2 = 13.33 + j5.33$$

$$\therefore V_{th} = V_2 = 14.36 \angle 21.80^\circ \text{ V}$$

Ans.

2) To find  $\bar{Z}_{th}$ , we consider the following circuit:



$$\bar{Z}_{th} = -j4 \parallel (4 + 10 \parallel j5) = 2.67 - j4 \Omega$$

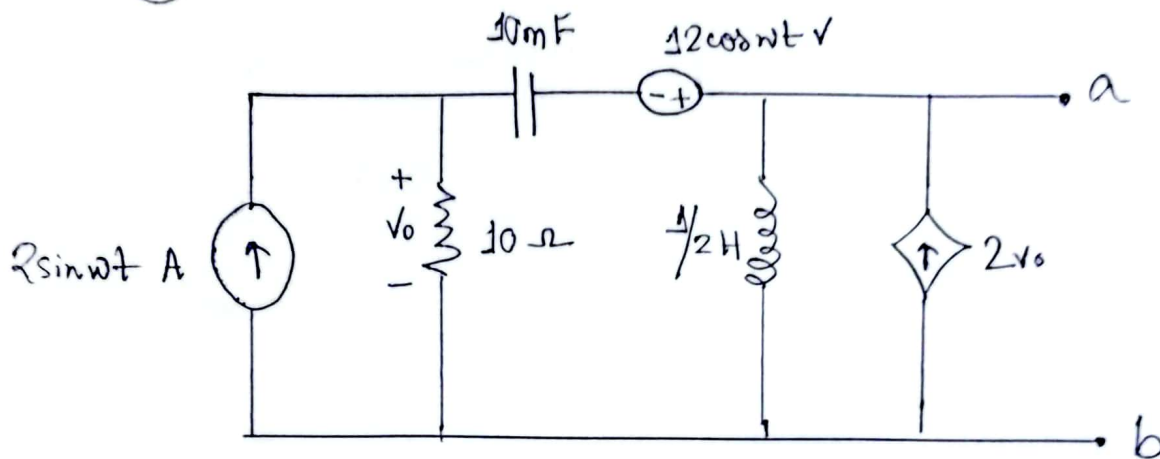
To find  $\bar{V}_{th}$ , we can use the calculation of (a).

$$\begin{aligned} \bar{V}_2 &= 13.33 + j5.33 \\ \bar{V}_1 &= (1-j)\bar{V}_2 + j32 \\ &= (1-j)(13.33 + j5.33) + j32 \\ &= 18.66 + j24 \end{aligned}$$

$$\therefore \bar{V}_{th} = \bar{V}_1 - \bar{V}_2 = 19.42 \angle 74.07^\circ \text{ V}$$

Ans.

o. At terminals a-b, obtain Thevenin and Norton equivalent circuits for the network depicted:  
 Given,  $\omega = 10 \text{ rad/s}$ .

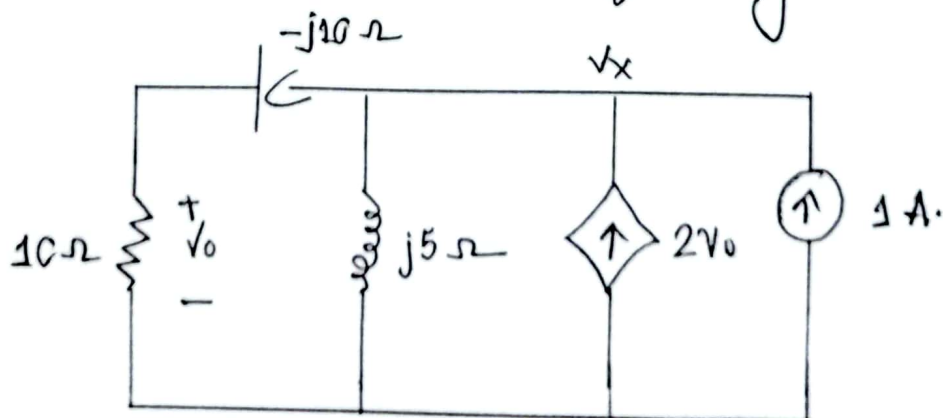


Solution: Given,  $\omega = 10 \text{ rad/s}$ .

$$0.5 \text{ H} \rightarrow j\omega L = j \times 10 \times 0.5 = j5$$

$$10 \text{ mF} \rightarrow \frac{1}{j\omega C} = \frac{1}{j \times 10 \times 10^{-3} \times 10} = -j10$$

To find  $Z_{th}$ , we consider the following circuit:



$$1 + 2V_o = \frac{V_x}{j5} + \frac{V_x}{10 - j10} \quad \text{--- (i)}$$

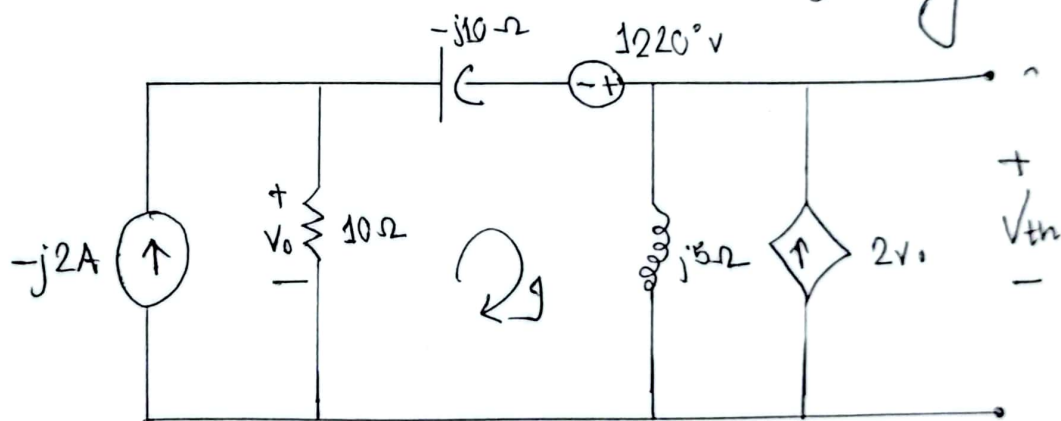
$$V_o = \frac{10V_x}{10 - j10} \quad \text{--- (ii)}$$

From (i) and (ii), we get:

$$V_x = \frac{-10 + j10}{21 + j2} = -0.43 + j0.51$$

$$\therefore Z_{in} = Z_{th} = \frac{V_x}{1} = 0.67 \angle 129.56^\circ$$

To find  $V_{th}$  and  $I_N$ , reconsider the following circuit:



$$(10 - j10 + j5) I - (10)(-j2) + j5(2V_o) = 12$$

$$\text{where, } V_o = 10(-j2 - I)$$

$$\text{Thus, } (10 - j105) I + 188 + j20 = 0$$

$$\Rightarrow I = \frac{188 + j20}{-10 + j105}$$



$$V_{th} = j5(1 + 20)$$

$$= j5(1 - j40 - 20j)$$

$$= -j95 + 200$$

$$= -j95 \left( \frac{188 + j20}{-10 + j105} \right) + 200$$

$$\Rightarrow V_{th} = 29.79 \angle -36^\circ \text{ V}$$

$$\therefore I_N = \frac{V_{th}}{Z_{th}} = \frac{29.79 \angle -36^\circ}{0.67 \angle 129.56^\circ} = 44.46 \angle -133.18^\circ$$

Ans.